VEHICLE RENTAL SYSTEM

A C++ Object-Oriented Programming Mini Project

Submitted by	Sneha
Roll Number	2556502
Course / Branch	B.Tech – Computer Science and Engineering
Institution	CGC Jhanjheri
Submitted to	Miss Yamini Sharma

PROJECT REPORT

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1. Introduction

The Vehicle Rental System is a menu-driven C++ console application that demonstrates Object-Oriented Programming (OOP) concepts. A user selects a vehicle category (Car or Bike), enters the vehicle information and rental duration, and receives a rental receipt showing the vehicle details and calculated charges.

A common base class stores shared vehicle information, while derived classes implement category-specific rental calculations.

2. Objectives

• Represent common vehicle information using a base class.

• Create Car and Bike classes using single inheritance.

• Calculate rental charges for different vehicle categories.

• Demonstrate function overriding and runtime polymorphism.

• Provide a menu-driven interface and validate user input.

3. Tools and Technologies

• Programming language: C++

• Concepts: classes, objects, constructors, encapsulation, inheritance, virtual functions, overriding and polymorphism

• Development environment: Visual Studio Code / online C++ compiler

• Compiler requirement: C++17-compatible compiler

4. System Design

4.1 Vehicle – Base Class

The Vehicle class stores common data members such as vehicle ID, model name and rental rate per day. Its constructor initializes these values. Virtual functions such as calculateRent(), getVehicleType() and displayDetails() provide a common interface for derived classes.

4.2 Car – Derived Class

The Car class inherits from Vehicle. It overrides the rental calculation to include the car-specific charge (for example, an insurance fee) in addition to the daily rental amount.

4.3 Bike – Derived Class

The Bike class also inherits from Vehicle. It overrides the rental calculation using the bike rental rate and the number of rental days.

4.4 Program Flow

• Display the main menu: Car, Bike or Exit.

• Read and validate the user's vehicle selection.

• Accept the vehicle ID, model and rental duration.

• Create the selected derived-class object through a Vehicle base-class pointer.

• Display vehicle details, duration and total rental charges.

• Release dynamically allocated memory and return to the menu.

5. Object-Oriented Programming Concepts

Classes and Objects

Vehicle, Car and Bike are classes. Objects represent individual vehicles created during program execution.

Encapsulation

Vehicle data members are kept protected and accessed through the class interface and member functions.

Inheritance

Car and Bike publicly inherit the common properties and behaviour of Vehicle. This is an example of single inheritance.

Function Overriding

Derived classes provide their own implementations of virtual functions declared in the base class.

Runtime Polymorphism

A Vehicle pointer can refer to a Car or Bike object. Calling an overridden virtual function through that pointer selects the correct derived-class implementation at runtime.

6. Input Validation and Testing

The program should accept valid menu choices, positive vehicle IDs and a positive rental duration. Invalid choices or non-positive values should be rejected and the user should be prompted again.

Test case	Input / action	Expected result
Car rental	Select Car; enter valid details and days	Car receipt and car rental total
Bike rental	Select Bike; enter valid details and days	Bike receipt and bike rental total
Invalid menu	Enter a choice outside the menu	Show an error or ask again
Invalid duration	Enter zero or a negative number	Reject the duration
Exit	Select Exit	Program terminates

7. Sample Rental Calculation

Illustrative example: if a bike has a daily rental rate of Rs. 500 and is rented for 3 days, the rental amount is Rs. 500 × 3 = Rs. 1,500. The actual receipt depends on the values entered by the user and the calculation implemented in the source code.

Illustrative car example: if a car has a daily rate of Rs. 1,500, is rented for 3 days and the program applies a fixed insurance fee of Rs. 200, the total is (Rs. 1,500 × 3) + Rs. 200 = Rs. 4,700.

8. Conclusion

The Vehicle Rental System demonstrates how C++ OOP features can be used to model a simple real-world rental process. Inheritance reduces repetition, virtual functions support runtime polymorphism, and overriding allows Car and Bike to apply different rental calculations. The project also provides practice in constructors, data organization, input handling and formatted output.

9. Source Code and Repository

The complete C++ source code, README file, sample output and this project report are maintained in the public GitHub repository:

https://github.com/snehasharma0205/Vehicle-Rental-System

Before final submission, compile and run the program, verify the calculations and attach screenshots of actual Car and Bike runs if required by the assignment.